

Generation Interconnection Request System Impact Study Report

For Queue Project Q22-2166

Bluestone 100 kV
January 22, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2166. The Interconnection Customer proposes a 179.8 MW solar photovoltaic facility interconnecting at the Bluestone 100 kV substation in the Illinois area. The request was received on August 22, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 618 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,343,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Cayuga – Fogarty 100kV	Estill Springs 230-1 kV Auto	109.0	157	13.3
Comed Devon – Fogarty 100kV	Estill Springs 230 kV/1 kV Autotransformer	107.8	129	13.3
North Branch Mn – Sedro Woolley 100kV Ckt 2	N Branc Mn – Sedro Woolley 100kV Line 1	105.8	144	9.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Cayuga – Fogarty 100kV	Reconductor line	\$393,000	83% (\$324,000)
Comed Devon – Fogarty 100kV	Reconductor line	\$683,000	83% (\$564,000)
North Branch Mn – Sedro Woolley 100kV Ckt 2	Reconductor line	\$455,000	100% (\$455,000)
	Total allocated to project		\$1,343,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.